

ANDREW AQUILINA

◇ LLM safety, alignment & evaluation · human-AI interaction · computational social science ◇

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Education

University of Pittsburgh
Ph.D. Information Science

Pittsburgh, PA
Aug 2024 – Apr 2029 (expected)

- Advisor: Prof. Yu-Ru Lin.
- **Research:** how socially-aware NLP systems assess and affect users' mental well-being.
- **Built:** multi-turn conversation simulators; LLM-as-judge and taxonomy-based response coding; linear probes over hidden states; an annotation study with 321 annotators and per-community labels.

Stockholm University
M.Sc. Computer and System Sciences, Artificial Intelligence

Stockholm, Sweden
Aug 2021 – Jun 2022

University of Malta
B.Sc. Information Technology, Artificial Intelligence

Msida, Malta
Oct 2018 – Jun 2021

- First Class Honours.

Publications

* Equal contribution.

Conference Papers

-  **A. Aquilina**, X. L. Li, and Y.-R. Lin. Whose Assessment of Distress? Community Perspectives and LLM Alignment on Well-Being Posts. *21st International AAAI Conference on Web and Social Media*, 2027. To appear. [\[code\]](#)
-  **A. Aquilina**, C. Nihalani, V. Varadarajan, N. S. Fishbein, Y.-R. Lin, and M. Sap. Lost in Delusion: Examining LLM Safety Under User Delusions and Distress. *Conference on Empirical Methods in Natural Language Processing*, 2026. Selected for oral presentation (17% of main-conference papers); to appear. [\[code\]](#)
-  **A. Aquilina** and C. Abela. Predicting Links in a Social Network Based on Recognised Personalities. *37th ACM/SIGAPP Symposium on Applied Computing*, 2022. [\[code\]](#)

Workshop Papers

-  M. Star*, **A. Aquilina***, and Y.-R. Lin. Auditing Support Strategies in LLMs through Grounded Multi-Turn Social Simulation. *SocialLLM Workshop at ICWSM*, 2026. Selected for oral presentation.

Preprints

-  **A. Aquilina**, S. Diacono, P. Papapetrou, and M. Movin. An End-to-End Workflow Using Topic Segmentation and Text Summarisation Methods for Improved Podcast Comprehension. arXiv:2307.13394, 2023.

Industry Experience

Loqus Business Intelligence
AI Engineer

San Ġwann, Malta
Sep 2023 – Jul 2024

- Prototyped the backend architecture for an LLM-powered customer-facing chatbot, integrating FastAPI, RabbitMQ, MongoDB, and LangChain to support retrieval and orchestration across services.
- Researched and evaluated approaches for integrating LLM-based conversational systems with company data and delivery-related workflows.

Sambla Group
Data Scientist

Stockholm, Sweden
Nov 2022 – May 2023

- Analyzed large-scale Swedish-language SMS data using NLP methods, including clustering, topic modeling, and summarization, to identify recurring customer communication patterns and inform a cost-reduction strategy.
- Developed and evaluated ranking approaches for personalized loan-offer recommendations, treating lender selection as a user-item ranking problem.

Ascent
Junior Data Scientist

Luqa, Malta
Jul 2021 – Aug 2022

- Contributed to the development of an item recommendation system in collaboration with a high-profile client in the beverage industry.

University of Malta
Junior Software Engineer

Msida, Malta
Jul 2020 – Sep 2020

- Developed an interactive game editor using OpenCV and Unity within the Department of Artificial Intelligence.

Ascent
Junior Software Engineer

Luqa, Malta
Jul 2019 – Apr 2020

- Assisted with the development of internal web and mobile applications using .NET technologies.

Technical Skills

Languages	Python, R, SQL, JavaScript, C#/.NET
Machine learning	PyTorch, Hugging Face Transformers, scikit-learn; deep learning, transformers, generative AI; interpretability, linear probing of hidden representations; CNN-based computer vision (OpenCV)
NLP & retrieval	Topic modeling, clustering, summarization, learning to rank and ranking optimization, recommender systems, information retrieval, retrieval-augmented generation, graph and social network analysis
LLM evaluation	Agent-based multi-turn social simulation, LLM-as-judge and taxonomy-based response coding, persona-grounded safety and robustness evaluation, calibration analysis, synthetic conversation data, open-weight and frontier foundation models, LangChain, vLLM
Study design	Applied statistics, mixed-effects models, mixed-methods and user studies, perspectivist and crowdsourced annotation, inter-rater reliability, experimental design
Data & infra	GCP, Slurm, Docker, FastAPI, Flask, MongoDB, RabbitMQ

Awards & Fellowships

HAIL Graduate Student Awardee , University of Pittsburgh, \$1.2k	2026 – 2027
Tertiary Education Scholarship , Government of Malta, \$60k	2024 – 2029
School of Computing and Information Fellowship , University of Pittsburgh	2024 – 2025
Pathfinder Scholarship , Malta's Digital Innovation Authority, \$4k	2021 – 2022

Teaching & Mentoring

- **Teaching assistant**, University of Pittsburgh: Foundations of Artificial Intelligence (Fall 2025); Big Ideas in Computing and Information (Fall 2025); Foundations of Cloud Computing for Data Science (Summer 2025).
- **Undergraduate research mentoring**, University of Pittsburgh: Michelle Star (2025–2026), Katya Surendran (2026–), Sean Jacobs (2025), Aishwarya Krishnamoorthy (2025).

Service & Outreach

- **Reviewer**: ICWSM 2026; ICWSM 2025 (Posters, Demos & Datasets Track); NEJLT 2025; GeBNLP Workshop at ACL 2025; Pluralistic Alignment Workshop at NeurIPS 2024.
- **Doctoral Guild**, University of Pittsburgh: Officer (Spring 2025), President (2025–2026).
- **IAESTE**, University of Malta: Officer (2020–2021).
- **Invited panelist**, Panel on Responsible Technology, organized by Tech For Good, University of Pittsburgh (Feb 2025).